



Self-Repaying Lending

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Self-Repaying Lending Against MOR

About \$20M of stETH sits in Morpheus and pays the depositor who supplied it nothing they can spend. lux/liquid makes that stETH borrowable without a sale and lists MOR as collateral for the first time.

The deposit pays the treasury, not the depositor. Third-party depositors supply roughly \$20M of stETH to the Morpheus capital pool [verified]. The yield, on the order of 5%, routes to Aave v3 through DistributorV2.sol (@aave/core-v3 IPool, line 10) [verified] and accrues to the protocol treasury. Seventy-five percent of that treasury flow funds a buyback, about \$511K per year [verified]. The depositor is left holding a claim on MOR emissions and stETH they cannot spend without unwinding the position. MOR does about as little. It farms emissions or it trades; market cap is about \$17M and its collateral use is zero [verified]. Neither the collateral nor the token can be borrowed against without giving one up.

The fix: lux/liquid. lux/liquid (MIT) is a self-repaying lending engine on Lux. Deposit a yield-bearing asset into VaultV2 and mint a transferable 1:1 synthetic against it, up to the position's collateralization floor [verified repo]. The collateral keeps yielding; that yield is harvested and burned against the debt, so the balance owed falls on its own with no repayment to schedule. Capital does not sit in one venue. It routes through about a dozen risk-capped adapters, among them Aave, Compound, Lido, EtherFi, Pendle, Morpho, Yearn, Ethena, EigenLayer, Maker DSR, and native Lux strategies [verified repo]. One counterparty's failure is bounded by its cap, not by the whole book. List MOR as a supported deposit and it becomes borrowable collateral.

The outcome, quantified. Take the \$20M of stETH already deposited [verified] at a 50% collateralization floor, standard for yield-bearing collateral [illustrative]. That is about \$10M of spendable liquidity minted against deposits that produce \$0 spendable for the depositor today [illustrative], and the depositor keeps the staking yield and the MOR emissions. MOR gains its first non-speculative demand: listed at an initial cap of \$2M [illustrative], it moves from zero collateral use to borrowable, and the demand is sourced from lending rather than from the emission schedule. Yield diversifies from Aave-only near 5% [verified] across a dozen capped adapters, each classified and allocated under governance.

What Morpheus gets

- Its stETH becomes spendable without a sale, and MOR becomes borrowable collateral for the first time. Nothing in the Morpheus capital contracts has to change to list it.
- A worked, MIT-licensed engine whose yield is spread across a dozen capped adapters instead of one, so treasury income no longer rides a single counterparty.

Relevance, and first step. One market on Lux lists stETH, self-repaying today, and MOR, borrowable today. lux/liquid is built and it is the lending leg of a set: it pairs with the bank that custodies the collateral (ADOPT-05) and the DEX that prices the synthetic and MOR against it (ADOPT-06). Lux is the counterparty that already runs all three, so the depositor never leaves one book to use the next.

What this does not solve. Self-repayment is a property of yield-bearing collateral. MOR listed as bare collateral is borrowable, but a MOR position only retires itself if MOR is put to work in a yield strategy of its own. The synthetic holds its peg through the redemption schedule and collateral quality, not by decree.

ADOPT-04, full paper available. lux/liquid • VaultV2 and a dozen adapters. Morpheus citation checked against SmartContracts/contracts/capital-protocol/DistributorV2.sol. • research@lux.network